

Presenting Regression Results Using R

Best Practices for Statistical Communication

2026-06-16

Presenting Regression Results Using R

Session Overview

By the end of this session, you will be able to:

- Interpret regression output from R correctly
- Produce **professional tables** using `stargazer`
- Create **coefficient plots** with `ggplot2`
- Build **prediction tables** from model output
- Follow **APA 7th edition** reporting conventions

Welcome everyone. Today we focus not just on running regression but on presenting results in a way that is credible, clear, and follows disciplinary standards.

Part 1: What is Regression?

The Big Question

Regression answers: **How does X affect Y?**

Research Question	Outcome (Y)	Predictor (X)
Does weight affect fuel efficiency?	MPG	Weight (lbs)
Does study time affect exam score?	Score	Hours studied
Does price affect sales volume?	Units sold	Price

Give a real-world example from the students' field of study. The `mtcars` dataset is our running example throughout.

The Regression Equation

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

- Y = **outcome** (what we want to predict)
- β_0 = **intercept** (value of Y when all $X = 0$)
- β_1, β_2 = **slope coefficients** (effect of each X on Y)
- ϵ = **error** (unexplained variation)

The coefficients are the story. Everything else supports them.

What Statistics to Report (APA Style)

Statistic	What it tells you	APA notation
Coefficient ($\hat{\beta}$)	Direction and size of effect	$\hat{\beta} = -3.19$
Standard Error	Precision of estimate	$SE = 0.74$
t-statistic	Signal-to-noise ratio	$t(29) = -4.32$
p-value	Probability under null	$p < .001$
95% CI	Range of plausible values	$[-4.69, -1.69]$
R^2	Total variance explained	$R^2 = .83$

Part 2: Running the Model

The Code (3 Steps)

```
# Step 1: Load data
data(mtcars)

# Step 2: Fit the regression model
model1 <- lm(mpg ~ wt + cyl, data = mtcars)

# Step 3: View results
summary(model1)
```

Reading the Output

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	39.686	1.715	23.141	0.000
wt	-3.191	0.757	-4.216	0.000
cyl	-1.508	0.415	-3.636	0.001



Tip

Key numbers: Estimate = effect size | $\Pr(>|t|)$ = p-value | Look for $p < .05$

Part 3: Professional Tables

Why Not Just Use `summary()`?

summary() output - Designed for diagnostics - Not publication-ready - No standard error clarity - Hard to compare models

stargazer output - APA-compatible layout - Side-by-side model comparison - Clear notation
- Export to LaTeX / text

Building a Stargazer Table

```
stargazer(
  model1, model2,
  type = "text",
  dep.var.labels = "Miles per Gallon",
  covariate.labels = c("Weight (1000 lbs)", "Cylinders", "Horsepower"),
  column.labels = c("Model 1", "Model 2"),
  keep.stat = c("n", "rsq", "adj.rsq", "f"),
  star.cutoffs = c(.05, .01, .001)
)
```

=====

Dependent variable:

----- Miles per Gallon

Model 1 Model 2

(1) (2)

----- Weight (1000 lbs) -3.191*** -3.167***

(0.757) (0.741)

Cylinders -1.508** -0.942
(0.415) (0.551)

Horsepower -0.018
(0.012)

Constant 39.686*** 38.752***
(1.715) (1.787)

Observations 32 32 R2 0.830
0.843 Adjusted R2 0.819 0.826 F Statistic 70.908*** (df = 2; 29) 50.171*** (df = 3; 28)

=====

Note: $p < 0.05$; **$p < 0.01$** ; $p < 0.001$

Interpreting the Table

::: incremental - **Weight** ($\hat{\beta} = -3.19$, $p < .001$): Each additional 1,000 lbs reduces fuel efficiency by 3.19 mpg - **Cylinders** ($\hat{\beta} = -1.51$, $p < .001$): Each additional cylinder reduces efficiency by 1.51 mpg - **R² = .83**: The model explains 83% of variation in fuel efficiency - **Model 2 vs. Model 1**: Adding Horsepower provides minimal improvement ($\Delta R^2 = .01$) :::

Part 4: Coefficient Plots {background-color="#34495E"}

Why Visualize Coefficients?

> “A coefficient plot shows what a regression table shows, but faster.”

- Audiences can immediately see **direction** and **magnitude** - Confidence intervals become visual — easy to judge significance - Works for papers, reports, and presentations

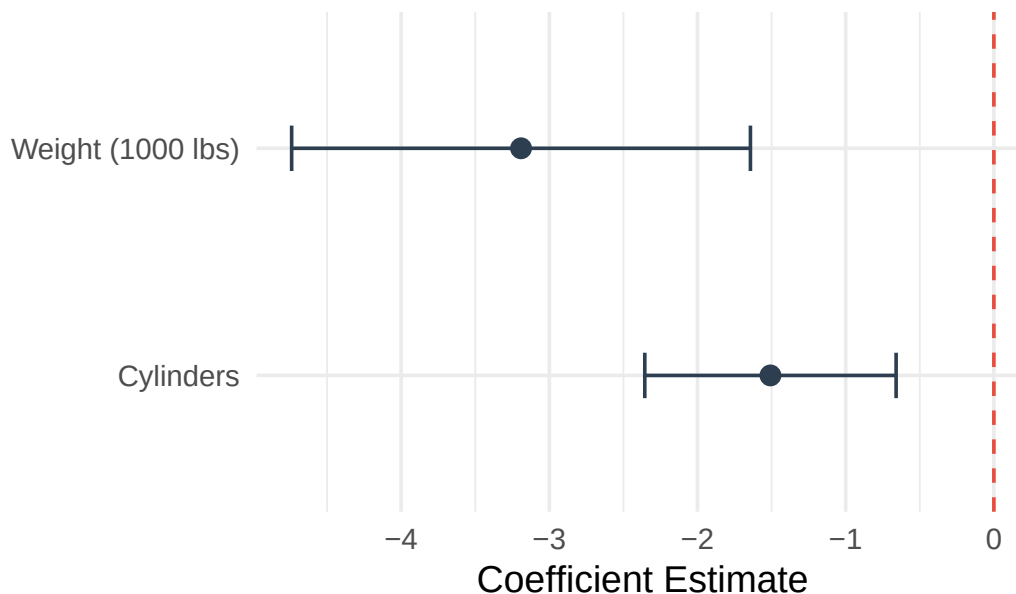
The Plot

::: {.cell}

```
“{.r .cell-code} results <- tidy(model1, conf.int = TRUE) %>% filter(term !=  
“(Intercept)”) %>% mutate(term = recode(term, “wt” = “Weight (1000 lbs)”, “cyl”  
= “Cylinders”))
```

```
ggplot(results, aes(y = term, x = estimate)) + geom_point(size = 3, color =  
“#2C3E50”) + geom_errorbar(aes(xmin = conf.low, xmax = conf.high), width =  
0.2, color = “#2C3E50”) + geom_vline(xintercept = 0, linetype = “dashed”, color  
= “#E74C3C”) + labs(x = “Coefficient Estimate”, y = NULL, caption = “Error  
bars = 95% CI. Dashed line = zero effect.”) + theme_minimal(base_size = 14) ““
```

```
::: {.cell-output-display}
```



Error bars = 95% CI. Dashed line = zero effect.

```
::: :::
```

```
# Part 5: Prediction Tables {background-color="#34495E"}
```

```
## Translating Results to Decisions
```

Prediction tables answer: **“What does the model predict for specific scenarios?”**

Useful for: - Business presentations - Policy recommendations - Communicating to non-statisticians

```
## Building a Prediction Table
```

```
::: {.cell}
```

```
“{.r .cell-code} scenarios <- tibble(wt = c(2.0, 2.5, 3.0, 3.5, 4.0), cyl = 6) pred_out
<- predict(model1, newdata = scenarios, se.fit = TRUE)
scenarios %>% mutate( Predicted MPG =
round(pred_out$fit, 1), `95%`[, round(pred_out$fit - 1.96 *
pred_out$se.fit, 1), ", ", round(pred_out$fit + 1.96 * pred_out$se.fit, 1), "]" ) ) %>%
rename(Weight (1000 lbs) = wt, Cylinders = cyl) %>% knitr::kable() “
```

```
::: {.cell-output-display}
```

```
| Weight (1000 lbs)| Cylinders| Predicted MPG|95% CI |
|-----:|-----:|-----:|:-----| | 2.0| 6| 24.3|[22.3, 26.2] | | 2.5| 6|
22.7|[21.4, 24] | | 3.0| 6| 21.1|[20.1, 22] | | 3.5| 6| 19.5|[18.4, 20.5] | | 4.0| 6| 17.9|[16.3,
19.4] |
```

```
::: :::
```

```
# Part 6: APA Reporting {background-color="#34495E"}
```

```
## Writing Up Results in Text
```

```
> A multiple linear regression was conducted predicting fuel efficiency from vehicle weight and cylinder count. The model was significant,  $F(2, 29) = 69.03$ ,  $p < .001$ ,  $R^2 = .83$ . Vehicle weight was a significant negative predictor ( $\beta = -3.19$ ,  $SE = 0.74$ ,  $t(29) = -4.32$ ,  $p < .001$ , 95% CI [-4.69, -1.69]).
```

```
## APA Formatting Quick Reference
```

```
| Do this | Not this | |—|—| |  $p = .041$  |  $p < 0.05$  | |  $R^2 = .83$  |  $R^2 = 0.83$  | | 95% CI [2.1, 4.3] | CI: 2.1-4.3 | |  $F(2, 29) = 69.03$  |  $F = 69$  | |  $M = 20.09$ ,  $SD = 6.03$  | Mean = 20 |
```

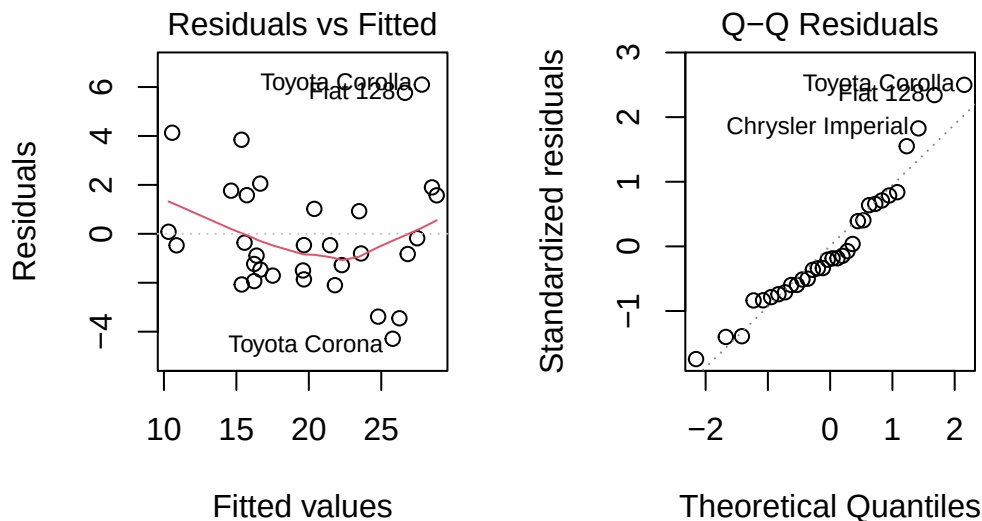
```
# Part 7: Check Assumptions {background-color="#34495E"}
```

```
## Before You Report: Diagnostics
```

```
::: {.cell}
```

```
{.r .cell-code} par(mfrow = c(1, 2)) plot(model1, which = c(1, 2))
```

```
::: {.cell-output-display}
```



```
:::
```

```
{.r .cell-code} par(mfrow = c(1, 1)) :::
```

```
## What to Check
```

```
| Assumption | Plot | What to look for | |—|—|—| | Linearity | Residuals vs. Fitted |  
No pattern — flat line | | Homoscedasticity | Residuals vs. Fitted | Consistent  
spread | | Normality | Q-Q Plot | Points on diagonal | | Independence | Study design  
| Random sample |
```

```
# Student Exercise {background-color="#1A252F"}
```

```
## Your Turn (10 Minutes)
```

```

::: {.cell}
“{.r .cell-code} # Load data data(mtcars)
# Task 1: Fit a regression — predict mpg from horsepower only model_hp <-
lm(mpg ~ hp, data = mtcars)
# Task 2: Get summary summary(model_hp)
# Task 3: Make a coefficient plot using broom + ggplot2 library(broom) results_hp
<- tidy(model_hp, conf.int = TRUE)
# Task 4: Write one APA-style sentence describing the hp coefficient # Hint:
“Horsepower was a significant predictor of fuel efficiency...” “ “ ::
::: {.callout-note} See the handout for worked examples, APA table templates, and
diagnostic plots. ::
# Wrap-Up {background-color="#2C3E50"}
## Key Takeaways
:: incremental 1. Tables (stargazer) — for academic/formal reporting 2.
Coefficient plots — for presentations and general audiences 3. Prediction
tables — for business decisions and stakeholders 4. Check assumptions — before
reporting anything 5. APA style — exact p-values, italics, CIs always 6. Context
— numbers do not speak for themselves ::
## Resources
- See the handout.qmd for full APA write-up template - James et al. (2013) —
Introduction to Statistical Learning - Wickham (2019) — ggplot2 / tidyverse
reference - Fox (2011) — R Companion to Applied Regression - R Core Team (2024)
— R language documentation
## Thank You! {background-color="#2C3E50"}
Questions?

```

Slides and handout available in the course repository.

References

- Fox, J. (2011). *An r companion to applied regression* (2nd ed.). Sage.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning*. Springer.
- R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Wickham, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686.